Effects of dietary coconut oil, butter and safflower oil on plasma lipids, lipoproteins and lathosterol levels.

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OBJECTIVE: The aim of this present study was to determine plasma levels of lathosterol, lipids, lipoproteins and apolipoproteins during diets rich in butter, coconut fat and safflower oil. DESIGN: The study consisted of sequential six week periods of diets rich in butter, coconut fat then safflower oil and measurements were made at baseline and at week 4 in each diet period. SUBJECTS: Forty-one healthy Pacific island polynesians living in New Zealand participated in the trial. INTERVENTIONS: Subjects were supplied with some foods rich in the test fats and were given detailed dietary advice which was reinforced regularly. RESULTS: Plasma lathosterol concentration ($P < 0.001$), the ratio plasma lathosterol/cholesterol ($P=0.04$), low density lipoprotein (LDL) cholesterol ($P<0.001$) and apoB ($P<0.001$) levels were significantly different among the diets and were significantly lower during coconut and safflower oil diets compared with butter diets. Plasma total cholesterol, HDL cholesterol and apoA-levels were also significantly ($P<$ or $=0.001$) different among the diets and were not significantly different between buffer and coconut diets. CONCLUSIONS: These data suggest that cholesterol synthesis is lower during diets rich in coconut fat and safflower oil compared with diets rich in butter and might be associated with lower production rates of apoB-containing lipoproteins.